



Chhatri

Paytm merchant income cover · Build for India AI Hackathon, Track 2

How the money moves

Where the premium goes, how the risk pool pays, who is allowed to approve money — and what two monsoons of backtest say about whether the pool balances.

4 min

from trigger to money in the merchant's Paytm balance

0

documents the merchant has to send on an area claim

0.65

claims paid per rupee of premium, across both monsoons

WHAT IS IN HERE

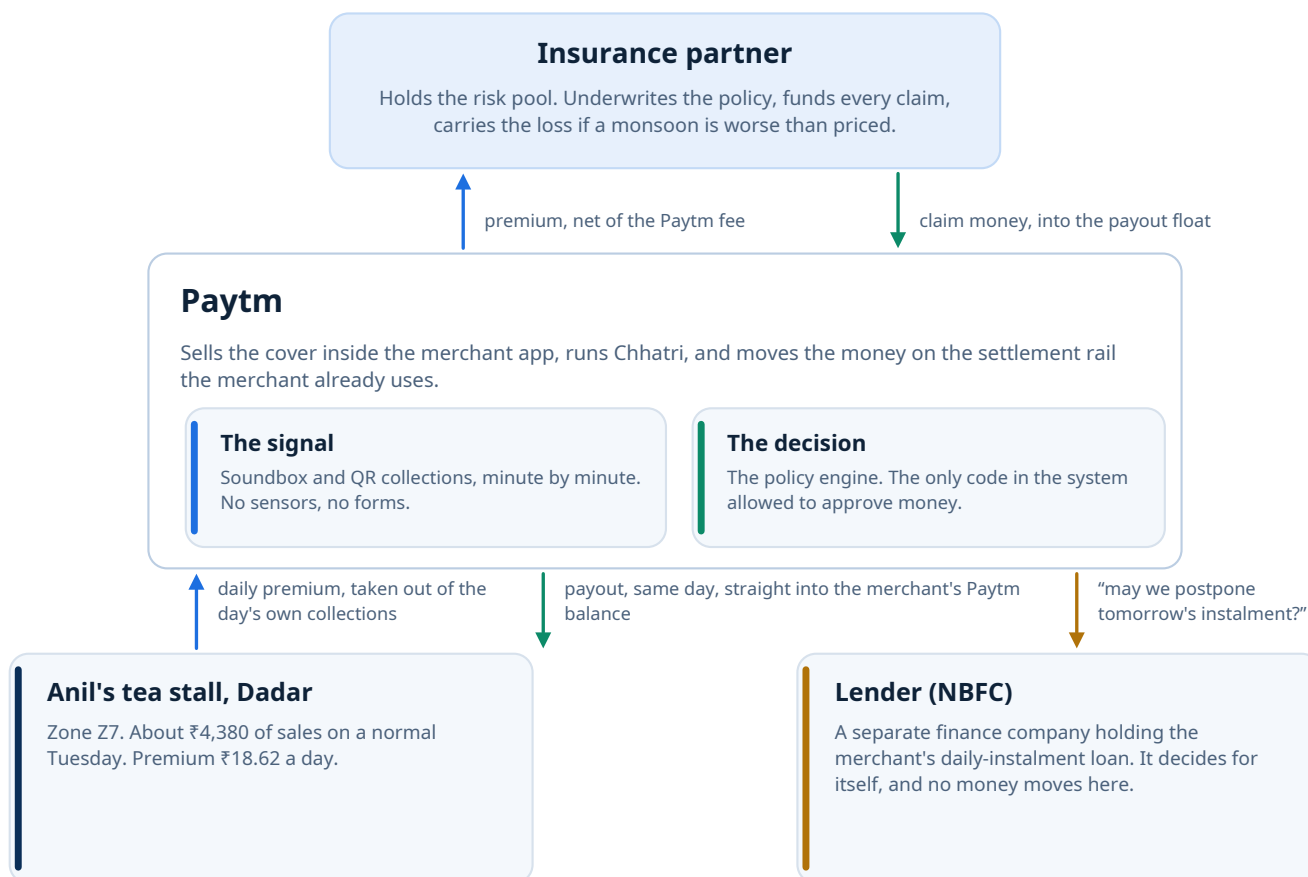
- 1 The big picture** who holds the risk, who moves the money
- 2 Money in** the premium, and the rule that no premium means no cover
- 3 Money out** the two ways a claim starts itself
- 4 The decision** eight hard rules, four soft ones, three outcomes
- 5 The storm** 14:00 to 17:05, with the arithmetic
- 6 Does the pool balance?** pricing, loss ratio and the backtest
- 7 Guardrails** limits, the audit chain, real versus simulated

The claim starts itself. The shops' own collections are the evidence, the policy engine is the only thing allowed to approve money, and the merchant hears the answer either way — in minutes, in their own language.

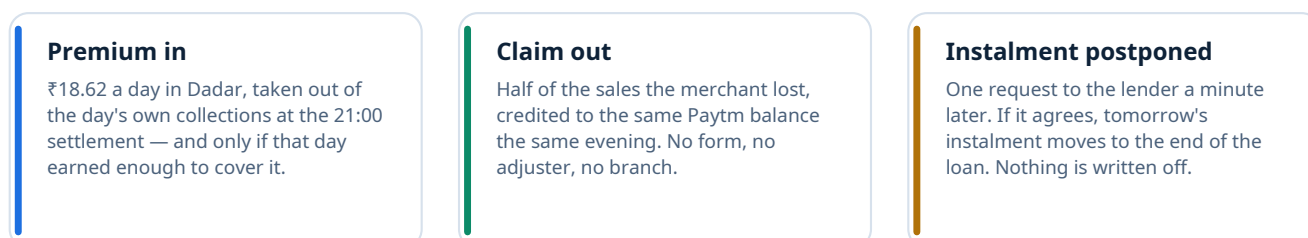
The whole product, in one sentence.

1 The big picture

Four parties, one rail. The merchant already takes money through Paytm every day, so Paytm is the only one of the four that can see the loss happen, collect the premium out of the takings, and push the payout back the same way. Chhatri is the piece in the middle that turns that stream of collections into a claim decision.

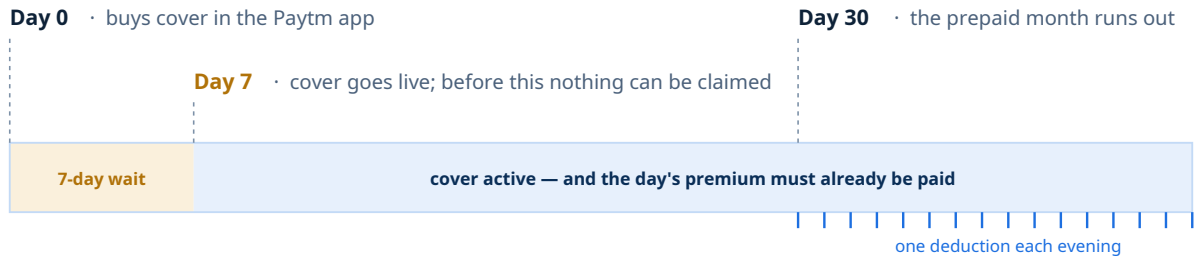


Paytm does not carry the risk. It distributes the cover and runs the engine; the insurance partner underwrites the policy and holds the pool. That split is what makes the product legal to sell and cheap to run — and it is why the engine, not a person and not a model, has the last word on whether money moves.



2 Money in — the premium

A merchant who cannot spare ₹500 on a bad week will not pay a yearly premium. So the premium is a daily number, charged the way the merchant already earns: out of the day's own collections, at the evening settlement.



Paid up front, once

A Paytm link for 30 days of cover: ₹18.62 × 30 = ₹558.60.
One tap, no mandate, no agent.

Then taken from the takings

At the 21:00 settlement Paytm deducts that day's ₹18.62 — but only if the day's collections cover it. A dead day costs the merchant nothing.

No premium, no cover

Section 64VB of the Insurance Act. The engine re-checks that the premium is prepaid through the date of the loss on every single claim, and declines if it is not.

A dead day costs nothing

If the day's collections do not cover the premium, nothing is deducted. The cover simply is not prepaid for the next day, and the merchant is told so.

No agent, no paperwork

Buying is two taps inside the app the merchant already uses to take payments. The 7-day wait is what stops anyone buying cover once the clouds appear.

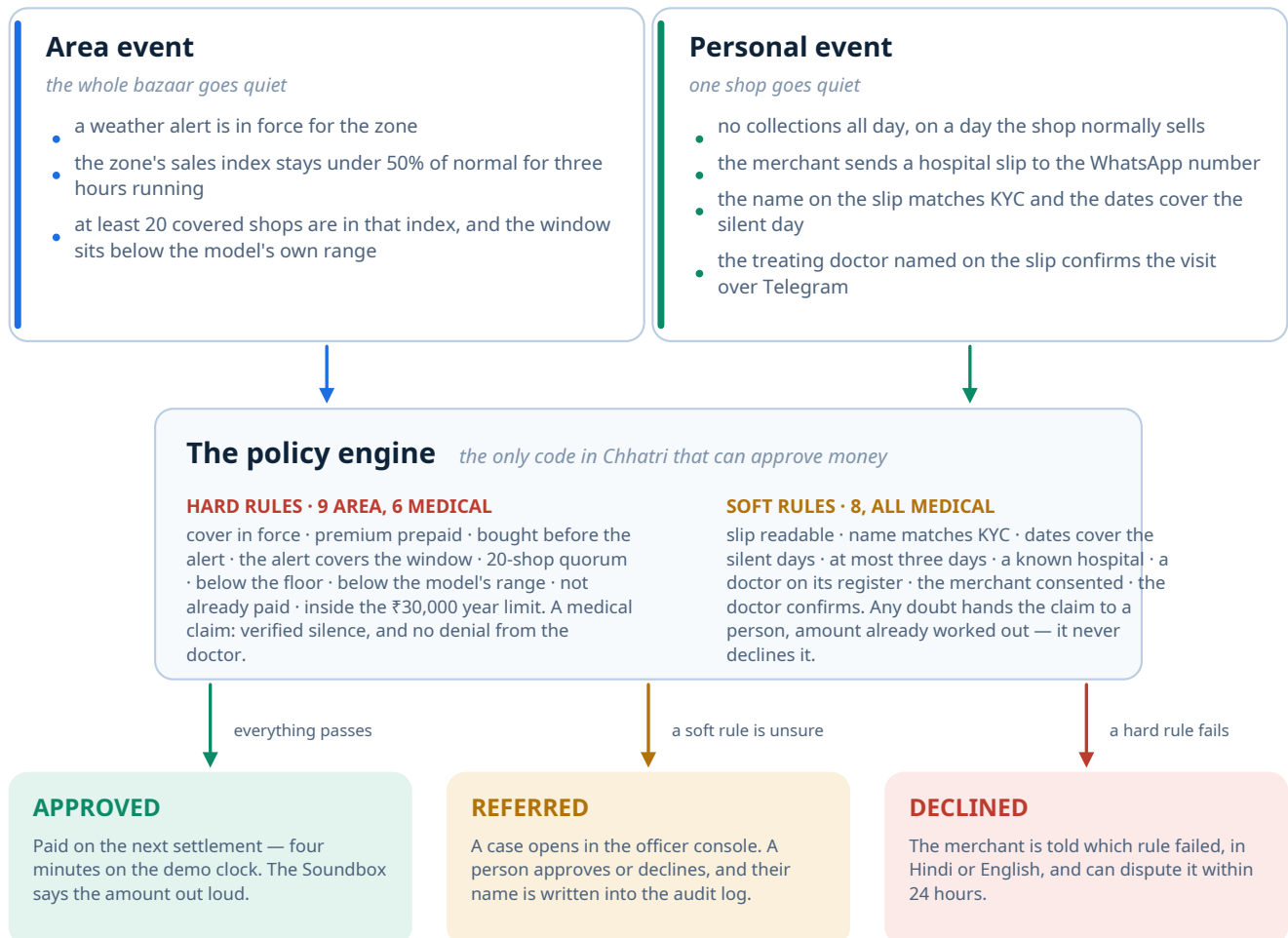
WHAT A DAY OF COVER COSTS, BY ZONE

per covered shop, priced off the backtest

Z8	₹38.82	Z13	₹16.25
Z6	₹29.84	Z17	₹15.98
Z20	₹28.79	Z3	₹14.16
Z9	₹25.38	Z2	₹13.71
Z14	₹21.36	Z19	₹13.66
Z18	₹21.14	Z1	₹13.55
Z24	₹19.59	Z11	₹10.89
Z22	₹19.06	Z12	₹10.48
Z7	₹18.62	Z5	₹9.87
Z15	₹17.04	Z16	₹9.34
Z4	₹16.84	Z23	₹7.19
Z10	₹16.60	Z21	₹6.93

3 Money out — the claim starts itself

There is no claim form. Two different silences start a claim: the whole market going quiet at once, which Chhatri can see without asking anyone, and one shop going quiet on its own, which needs one photograph and sometimes one human.



WHAT ANIL SEES ON THE STORM DAY

Anil ji, heavy rain cut your area's sales by 63% today.

₹1,380 — credited with today's settlement. No claim needed.

Your lender has paused tomorrow's instalment. It moves to the end of your loan with no penalty.

WHAT HE SEES WHEN HE IS THE ONE WHO IS ILL

Get well soon. Please send one photo of the hospital slip.

Anil ji, your claim is approved. ₹1,500 credited with today's settlement.

And if the name had not matched his KYC: our team will check it, and he hears back within 24 hours.

These are the merchant-facing strings from the message catalogue in docs/SPEC.md §13.4, with the amounts filled in. The merchant gets them in Hindi or in English, whichever they already use with Paytm.

4 The decision — what is actually checked

These are the rules in `policy/rules.yaml`, not a description of them. A hard rule that fails ends the claim and is named in the message the merchant gets. A soft rule that is merely unsure never declines anything — it sends the case to a person, with the amount already computed.

Rule	Applies to	Kind	Passes when
COVER_IN_FORCE	both	HARD	the cover exists, has started, and is active
PREMIUM_PREPAID	both	HARD	the premium is paid through the date of the loss
COVER_BEFORE_ALERT	area	HARD	the cover was bought before the alert was issued
ALERT_ACTIVE	area	HARD	the weather alert covers the whole trigger window
INDEX_QUORUM	area	HARD	at least 20 covered shops are in the zone index
BELOW_FLOOR	area	HARD	all three hourly readings are under 50% of normal
BELOW_MODEL_RANGE	area	HARD	the window is below the model's own lower bound
SILENCE_VERIFIED	personal	HARD	each claimed day really is a verified silent day
DOCTOR_NOT_DENIED	personal	HARD	the treating doctor has not said the patient was never there
NOT_ALREADY_PAID	both	HARD	nothing has been paid for that merchant, date and kind
WITHIN_ANNUAL_LIMIT	both	HARD	what is already paid in the rolling 365 days, plus this amount, stays under ₹30,000
SLIP_READABLE	personal	SOFT	a medical slip is attached and read with confidence 0.80 or better
NAME_MATCHES_KYC	personal	SOFT	the name on the slip scores 85 or more against the KYC name
DATES_MATCH	personal	SOFT	admission and discharge cover every silent day claimed
HOSPITAL_IDENTIFIED	personal	SOFT	the slip names a hospital Chhatri's own directory knows
DOCTOR_IDENTIFIED	personal	SOFT	it names a doctor on that hospital's register
VERIFICATION_CONSENT	personal	SOFT	the merchant agreed to their doctor being asked
DOCTOR_CONFIRMED	personal	SOFT	that doctor confirms the visit; silence refers, never declines
WITHIN_AUTO_LIMIT	personal	SOFT	three silent days or fewer; anything longer goes to a person

Caps sit on top of the rules: an area day is capped at ₹2,500, a personal day at ₹1,500, and every merchant has a ₹30,000 ceiling across a rolling 365 days — rolling, so there is no policy-year boundary to game.

When a rule fails

The merchant is told which one, in plain words, and can argue with it in the same chat. A dispute opens a case within 24 hours.

When a person decides

The officer sees exactly the facts the engine saw, approves or declines, and their name and reason go into the log.

Nothing is invisible

Every decision, payout and override is hashed into a chain. Change one row after the fact and the chain stops verifying.

5 The storm, minute by minute

This is the demo. One zone, one afternoon, and nobody filing anything. The times below are the simulated clock the replay runs on, so the same storm lands on the same numbers every time it is played.

- 14:00** **The alert is issued**
 IMD puts a heavy-rain warning over the Dadar cluster and Chhatri arms the zone. From this moment on, cover bought today cannot claim on today's storm.
- 16:00** **The rain lands**
 QR and Soundbox collections across Z7 fall away. Nobody files anything — the collections themselves are the evidence.
- 17:00** **The third hour closes below the floor**
 The Z7 index reads 37% of normal across 46 covered shops: under the 50% floor for three hours running, and under the model's own lower bound. The trigger fires.
- 17:00** **The engine decides, shop by shop**
 Eight hard rules per merchant, no documents, no human in the way. 312 shops across Z7, Z3 and Z12 are approved; Z9 sits at 61% and is correctly left alone.
- 17:04** **The money lands**
 Anil receives ₹1,380. Z7 is paid ₹58,900 in all. The Soundbox announces the credit out loud, so he hears it without opening the app.
- 17:05** **The lender is asked**
 Chhatri asks Anil's NBFC to postpone tomorrow's ₹600 instalment to the end of the loan. The lender decides on its own rules; whatever it answers, the payout stands.

The storm payout

what Anil actually gets on 19 August

expected day, published to the nearest ₹10	₹4,380
zone index 37% of normal, so the drop is	63%
sales lost that day	₹2,759.40
Chhatri pays half of the loss	₹1,379.70
rounded to the rupee	₹1,380
cap on an area day — not reached	₹2,500

CREDITED AT 17:04

₹1,380

The hospital payout

the same shop, silent for one day in September

expected day, published	₹4,380
half of the expected day	₹2,190
cap on a personal day — this one bites	₹1,500
one verified silent day	× 1
slip read, name scored against KYC	pass
KEM Hospital, Dr S. Rao confirmed on Telegram	pass
over three days it would go to a person	—

CREDITED AFTER THE SLIP

₹1,500

6 Does the pool balance?

A cover that pays too rarely is not worth buying, and one that pays too easily cannot be sold twice. The backtest runs both triggers over two real monsoons of rainfall and asks the only question an underwriter cares about: what came in, and what went out.

What a merchant pays

$$\text{premium per day} = \max(₹2, \text{expected yearly loss} \div 365 \div (1 - 0.35))$$

The 0.35 is the loading — claims handling, reinsurance and margin. Priced per zone from the backtest: ₹6.93 to ₹38.82 a day across the 24 Mumbai zones; Dadar (Z7) is ₹18.62.

WHERE EVERY ₹100 OF PREMIUM GOES

₹65 goes back out as claims

₹35 loading

	Chhattri	Rain-only trigger
What it watches	the shops' own collections, hour by hour	millimetres of rain at a grid point
Real bad days it caught	89 of 148 · 60%	49 of 148 · 33%
Payouts where nothing really went wrong	36 of 125 · 29%	287 of 336 · 85%
Paid out over the two monsoons	₹1.49 crore	₹3.36 crore
Against ₹2.30 crore of premium	loss ratio 0.65	loss ratio 1.46 — the pool breaks
Time from trigger to money	same day · four minutes	the next day, once the rain total is in
Documents asked of the merchant	none	none

Personal claims over the same two monsoons

844

medical claims in all

485

paid by the engine alone

359

sent to a person — 43% of them

7 Guardrails, and what is real

The fastest way to lose an insurer's licence is to pay money a rule did not allow. So everything that can move money is bounded, logged and replayable — and the parts that are simulated are named as simulated.

A hard limit, not a soft one

₹30,000 per merchant across a rolling 365 days, checked on every claim before the money moves. No policy-year reset to game.

Caps on each payout

An area day is capped at ₹2,500 and a personal day at ₹1,500, so a single bad model reading cannot drain the pool.

Humans keep the hard calls

Anything over three days of illness, an unreadable slip, a name that does not match, or a doctor who does not answer goes to an officer. In the backtest, which had no hospital to ask, that was 359 of 844 personal claims — 43%.

Every decision is replayable

Each decision, payout and officer action is hashed into a tamper-evident chain in the audit log. The whole storm can be replayed on a simulated clock and lands on the same numbers.

Real	Simulated
Rainfall: the actual hourly Open-Meteo record for Santacruz and Colaba, June–September 2024 and 2025.	Sales: 1,821 shops across 24 Mumbai zones, generated — Paytm's real collections are not public data.
The money rules, the policy engine, the tamper-evident audit chain and the deterministic replay are the real implementation, with tests.	The payout rail and the lender's instalment API sit behind interfaces and are stubbed. No real money moves anywhere.
The WhatsApp and Telegram journeys run against live bots, in Hindi and English.	The merchants, their KYC records and the hospital slips are fictional.

One honest caveat on price. Priced off simulated sales, the premium lands between ₹6.93 and ₹38.82 a day depending on the zone — ₹18.62 in Dadar. That is what this backtest supports. The “a few rupees a day” line in the pitch only holds for the cheapest zones, and real collections data would move every one of these numbers.

The three questions this usually gets

“Why not just pay on rainfall?”

Because rain is not loss. The rain-only trigger pays when nothing actually went wrong 85% of the time, and would pay ₹3.36 crore against ₹2.30 crore of premium — the pool breaks in a single season.

“Can a merchant game it?”

No single shop can move the index. It takes 20 covered shops, three hours under the floor and a live alert — and cover bought after the alert is issued is excluded.

“What if the model is wrong?”

A wrong model can only ever pay less. The window has to sit below the model's own range and below a fixed 50% floor, and an area day is capped at ₹2,500.

Figures in this document are read straight from docs/SPEC.md, backend/chhattri/policy/rules.yaml, backend/artifacts/premiums.json and backend/artifacts/backtest/report.json at the moment it was generated.